



GeoSciML JSON and Vocabularies

OGC Builder Days Code Sprint

Geovation, London, United Kingdom

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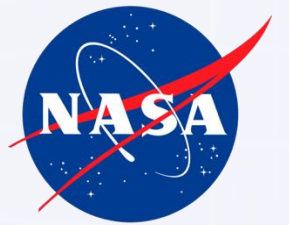
Clemens Portele | interactive instrument GmbH

11 May 2026

Meeting Host



Meeting Sponsor



Agenda

- Introduction
- Background and motivation
- GeoSciML Logical Model
- GeoSciML Basic requirements class
- How the JSON Schema files will be developed
- GeoSciML Vocabularies

Introduction

- The Geoscience Markup Language (GeoSciML) describes a logical model and GML/XML encoding rules for the exchange of geological map data, geological time scales, boreholes, and metadata for laboratory analyses.
- The standard provides models for concepts at the immediate periphery of geological mapping, such as boreholes, geologic specimens and laboratory analysis.
- Origins:
 - Originally developed by the Commission for the Management and Application of Geoscience Information (CGI)
 - Standardized by the Open Geospatial Consortium (OGC)

Timeline

2003: Initiation
under CGI WG
on Data Model
Collaboration

2012: GeoSciML
v3.1 Released

2017: GeoSciML
v4.1 Released

2006-2007:
OneGeology
Impact

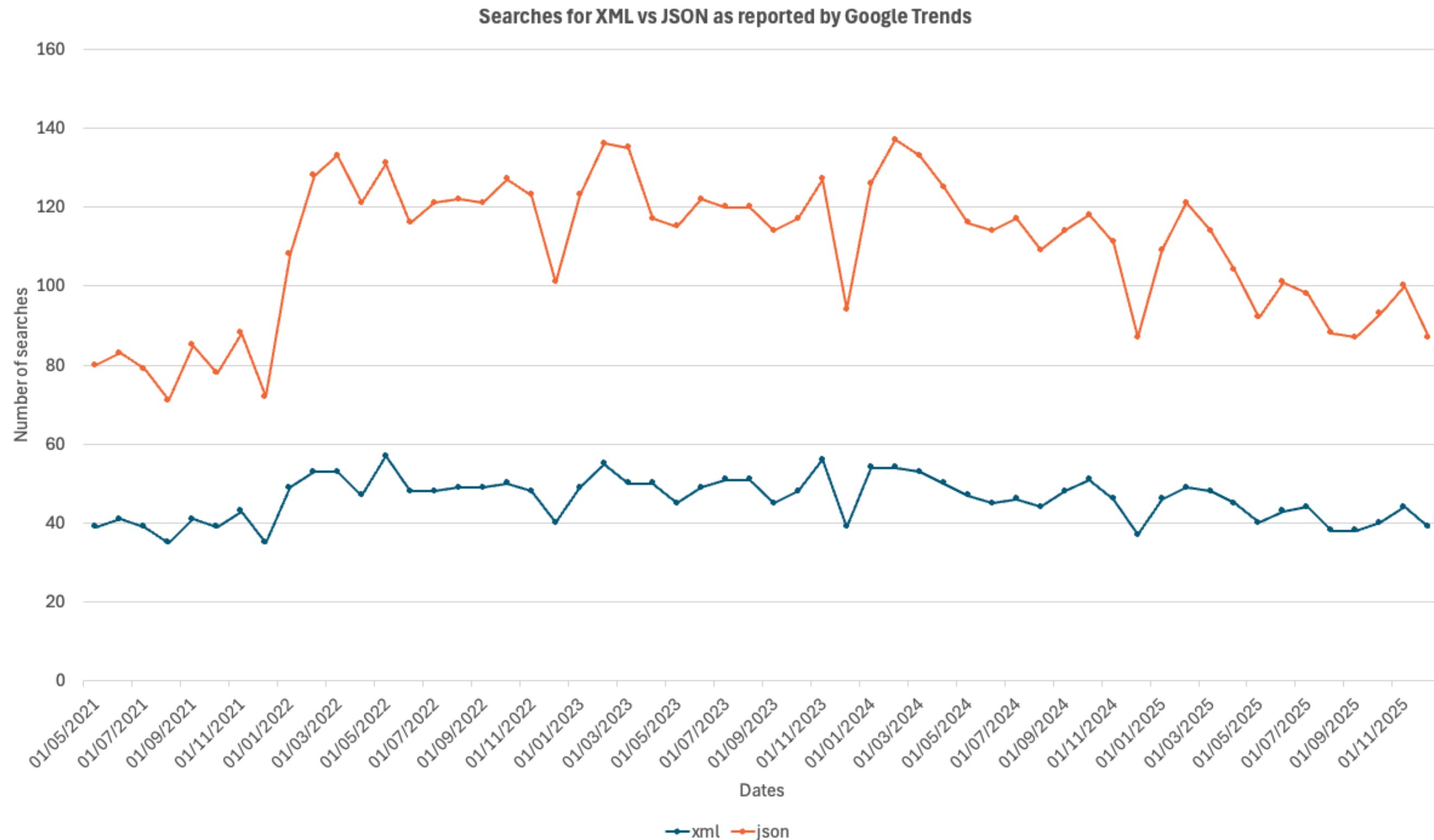
2013-2017:
OGC
Standardization:
OGC GeoSciML
Standards
Working Group
established

2017:
Discussions held
at 104th TC
Meeting about
alternative
encodings,
including
RDF/XML



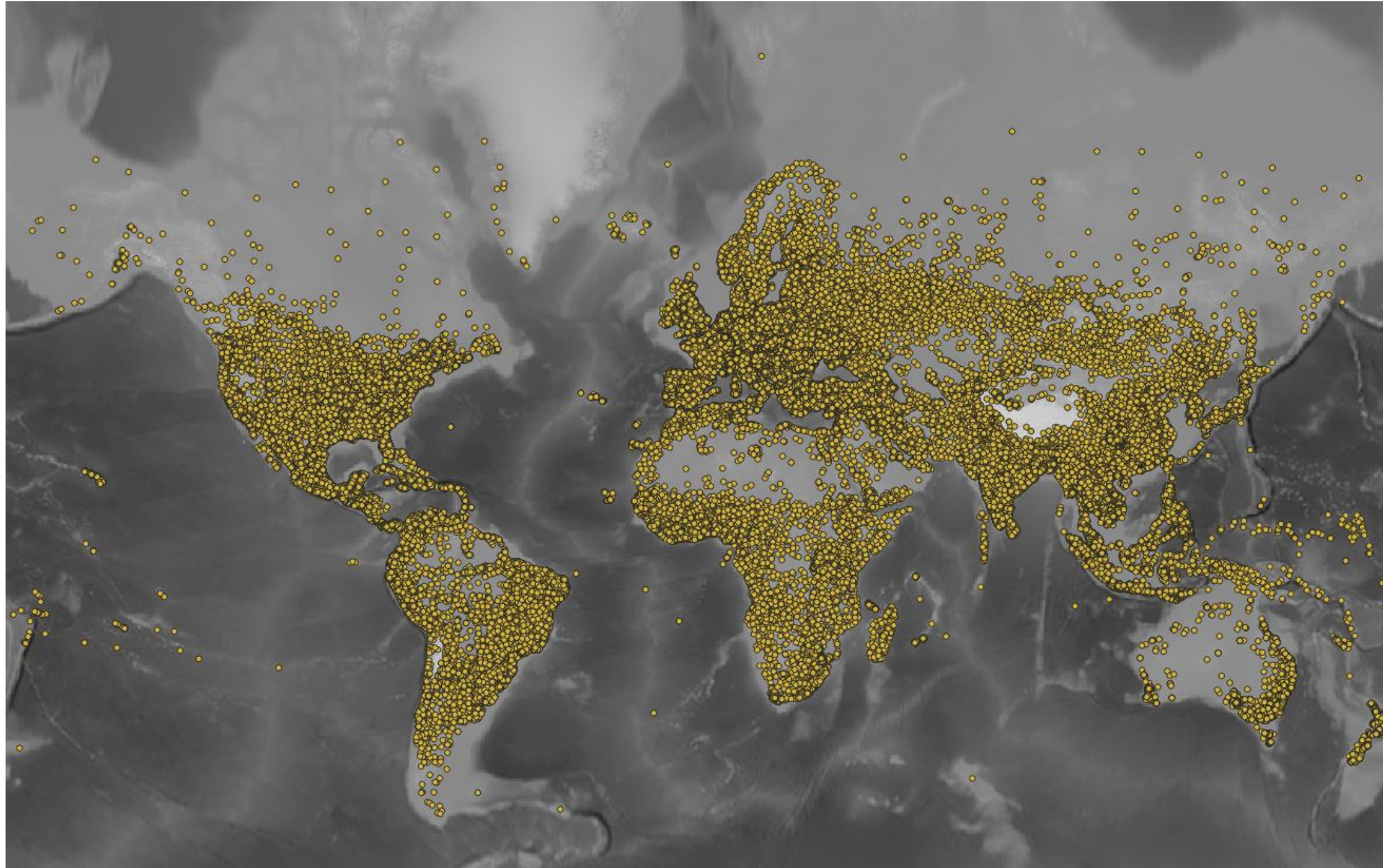
Why develop a JSON encoding for GeoSciML? (1)

- JSON increasingly becoming the most widely used message encoding on the Web



Why develop a JSON encoding for GeoSciML? (2)

- JSON is generally less verbose than XML for the same information



cities.gml

Today at 10:12

62.3 MB



cities.json

Today at 10:13

39.1 MB

Exported using
QGIS



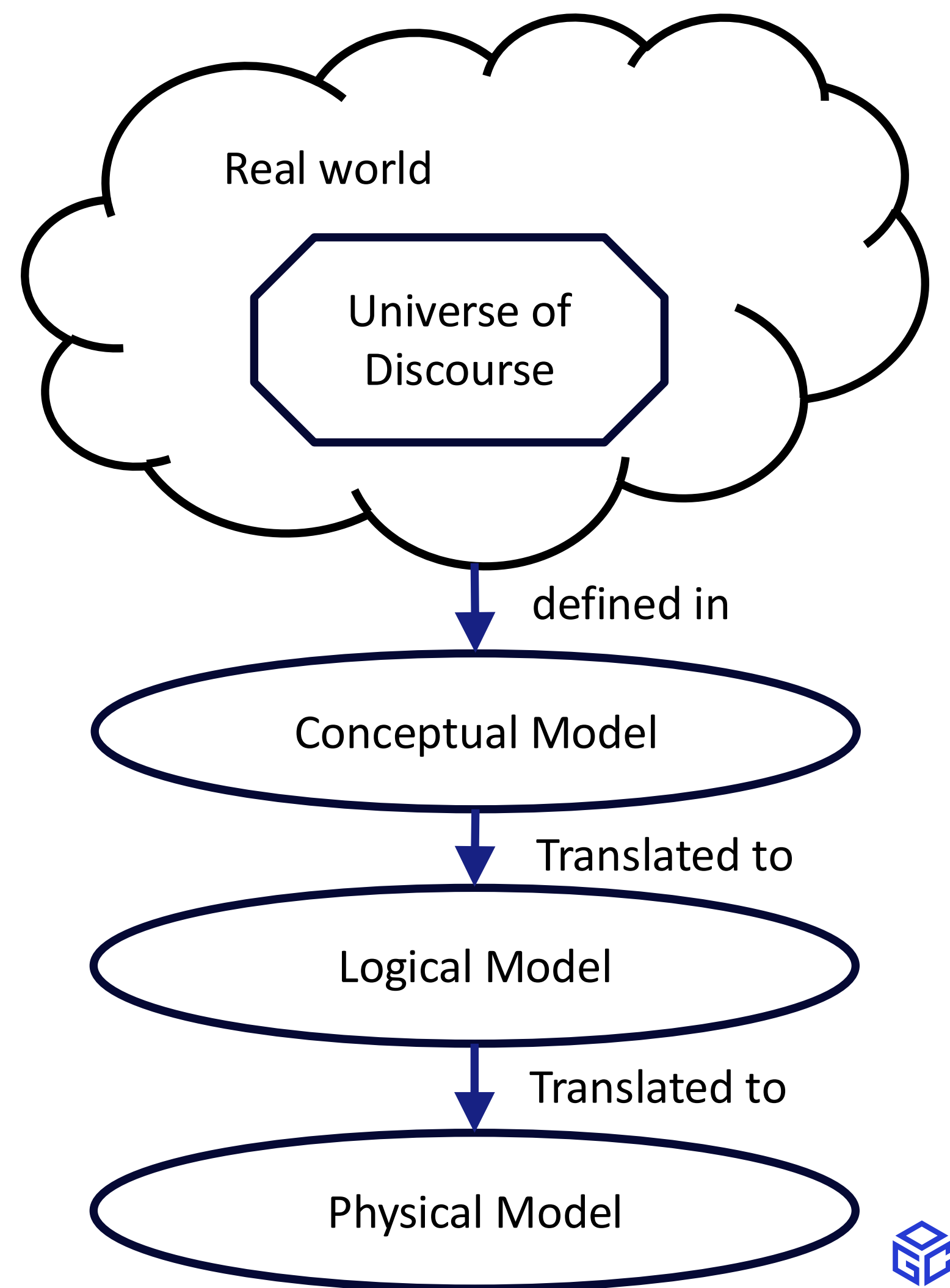
Why develop a JSON encoding for GeoSciML? (3)

- On 2026-05-09, asked three popular AI platforms the same question:
 - xml or json - which is better for AI platforms?
- Here are their responses
 - “Both have their place, but **JSON** wins for most AI platform use cases...”
 - “For most modern AI platforms (LLM APIs, agent frameworks, tool/function calling, RAG pipelines), **JSON** is the default and best fit because...”
 - “Both JSON and XML are valuable for AI platforms, but they excel in different areas. **JSON** is generally superior for speed, data interchange, and API communication, while **XML** is increasingly favored for structuring complex prompts in LLM applications due to”



Fundamentals of Data Modeling

- Conceptual Model
 - Represents high-level entities and their relationships
 - model that defines concepts of a universe of discourse (a view of the real world that includes everything of interest)
 - Technology-independent
- Logical Model
 - Translated from the CM to include additional details about the entities, their attributes and relationships
 - Technology-independent
- Physical Model
 - Translated from the LM to include specific details about physical encodings (e.g. GML/XML or JSON)
 - Technology specific



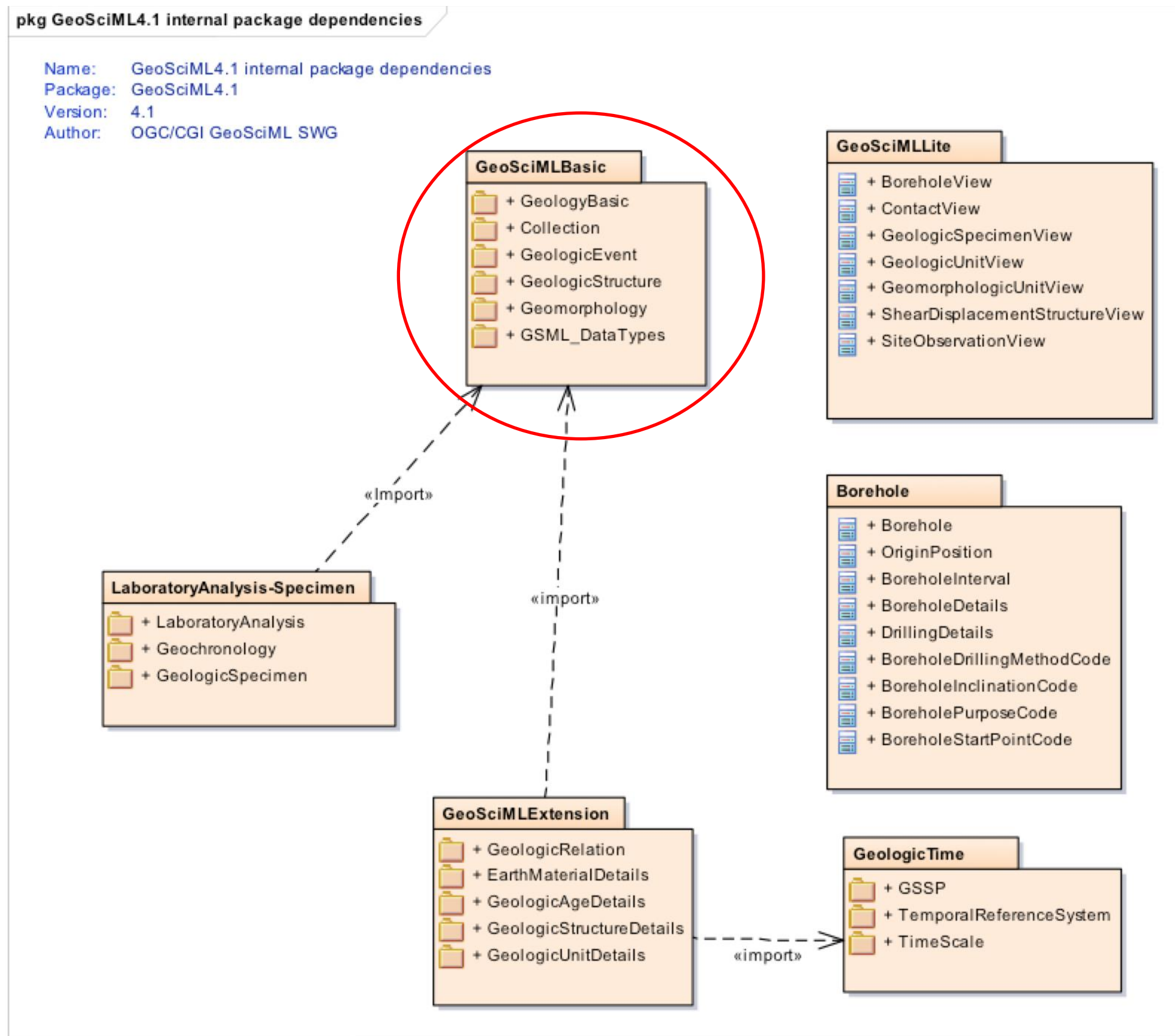
Packages of the GeoSciML Logical Model

- Organized by themes into 6 packages organized by use cases:

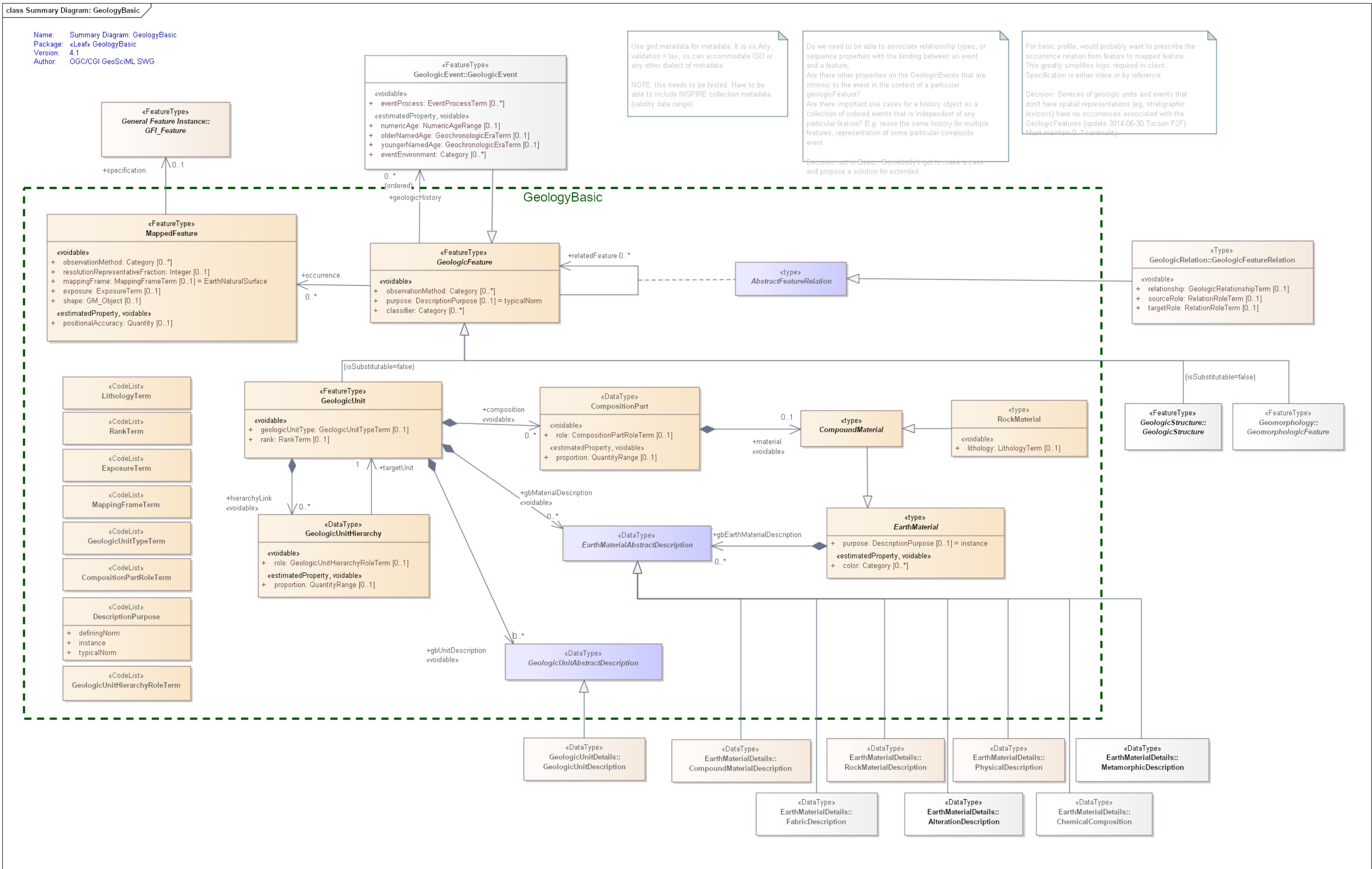
Sprint
Focus

- **GeoSciML Basic:** a set of core geologic features, aligned to the INSPIRE Data Specification on Geology.
- **GeoSciML Extension:** an extension providing detailed description of basic features which adds additional properties and associations.
- **GeoSciML Geologic Age:** a model for the representation of geologic time using procedures adopted by the International Stratigraphic Commission.
- **GeoSciML Borehole:** a model for boreholes, including geologic logs and drilling details and other engineering information.
- **GeoSciML Laboratory and Analysis:** a model for laboratory analytical metadata, geological sampling and specimens, and isotopic age observation results.
- **GeoSciML Lite:** previously known as “GeoSciML Portrayal” in version 3.2; a simplified alternate implementation of the conceptual model for layer based applications.

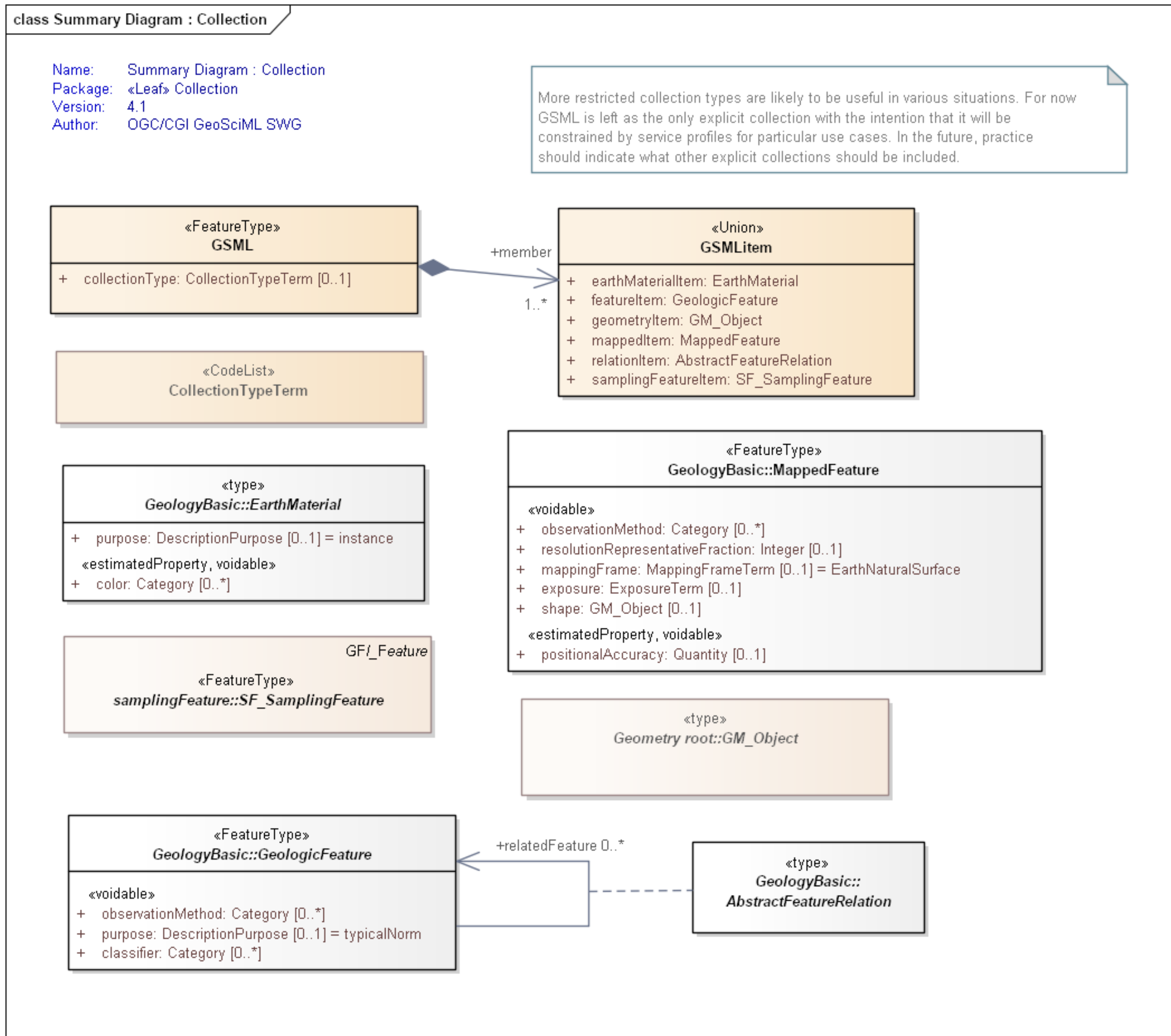
Package dependencies



GeologyBasic



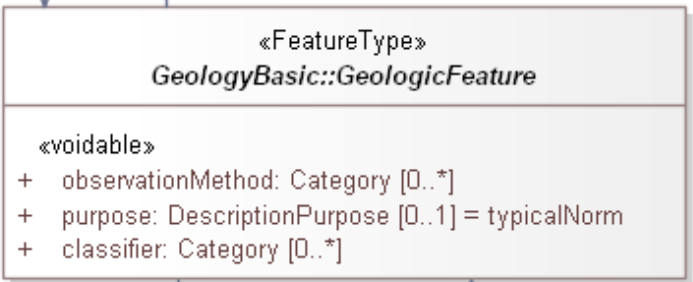
Collection



Name: Summary Diagram : GeologicEvent
Package: «leaf» GeologicEvent
Version: 4.1
Author: OGC/CGI GeoSciML SWG

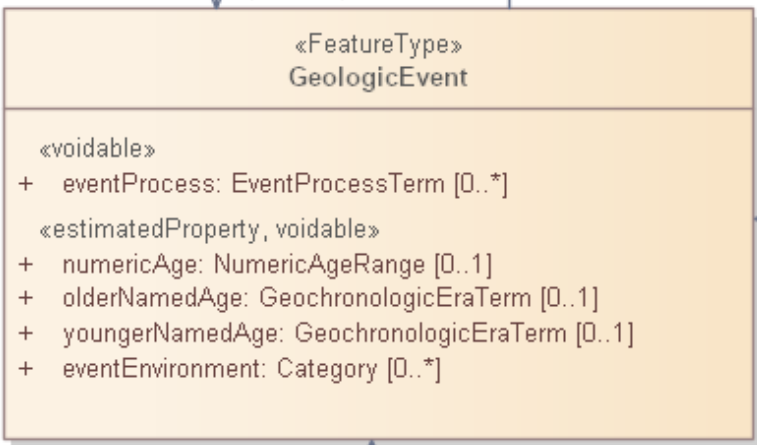
For relations between GeologicUnits/
Structures and GeologicEvents, use the
"geologicHistory" association, not the
generic "relatedfeature" association.

+relatedFeature 0..*

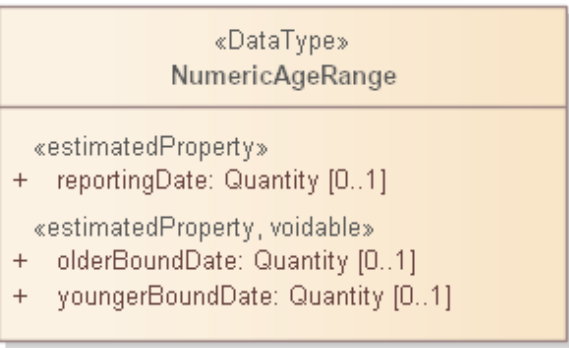


+geologicHistory

0..*
{ordered}



«FeatureType»
**GeologicStructureDetails::
DisplacementEvent**

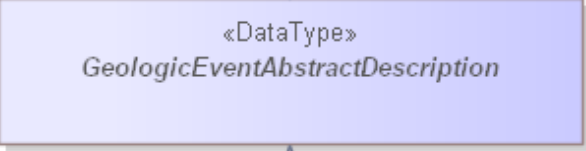


«CodeList»
GeochronologicEraTerm

«CodeList»
EventProcessTerm

+gaEventDescription

0..*



«DataType»
GeologicAgeDetails::GeologicEventDescription

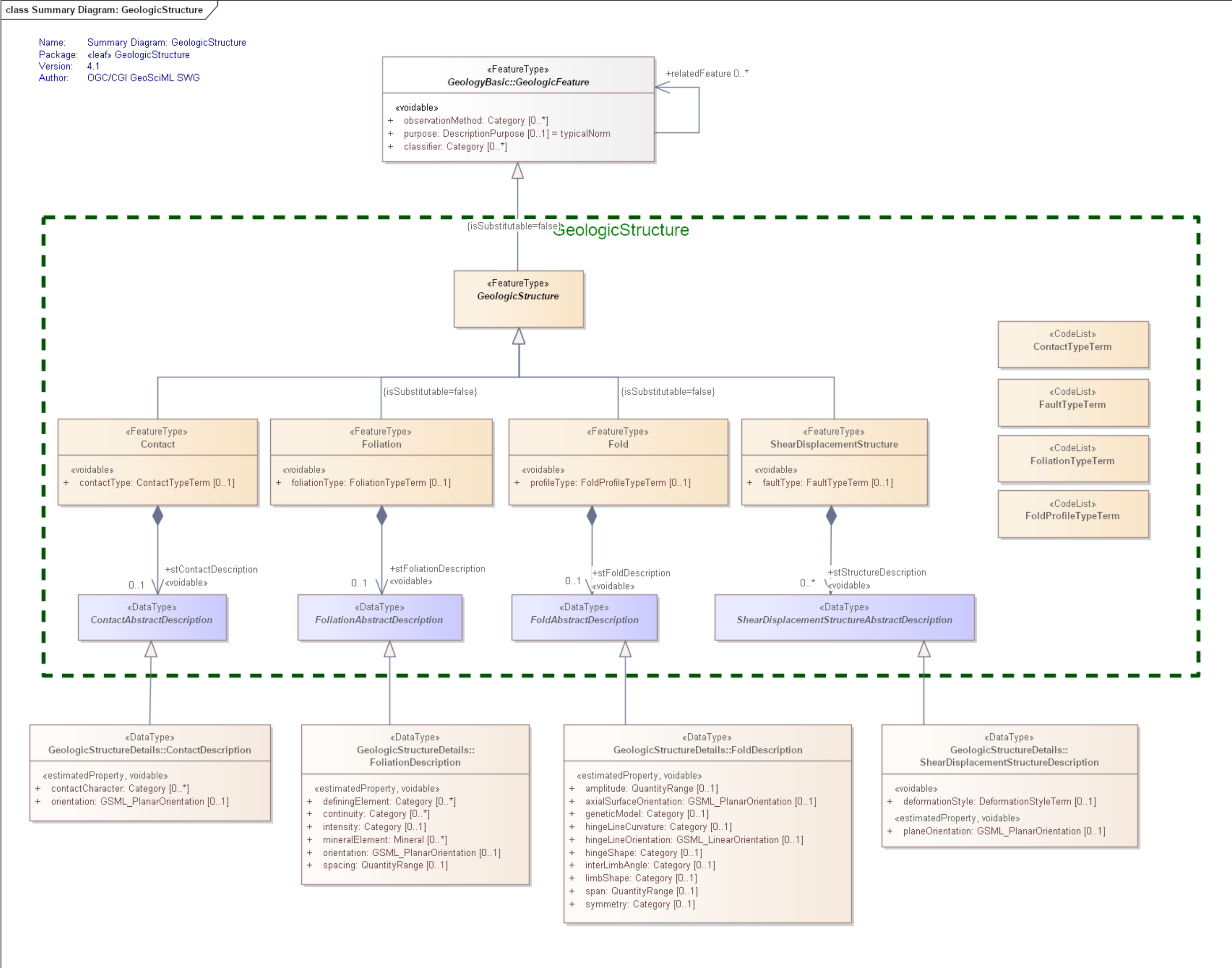
«voidable»
+ olderGeochronologicEra: GeochronologicEra [0..1]
+ youngerGeochronologicEra: GeochronologicEra [0..1]

GeologicEvent



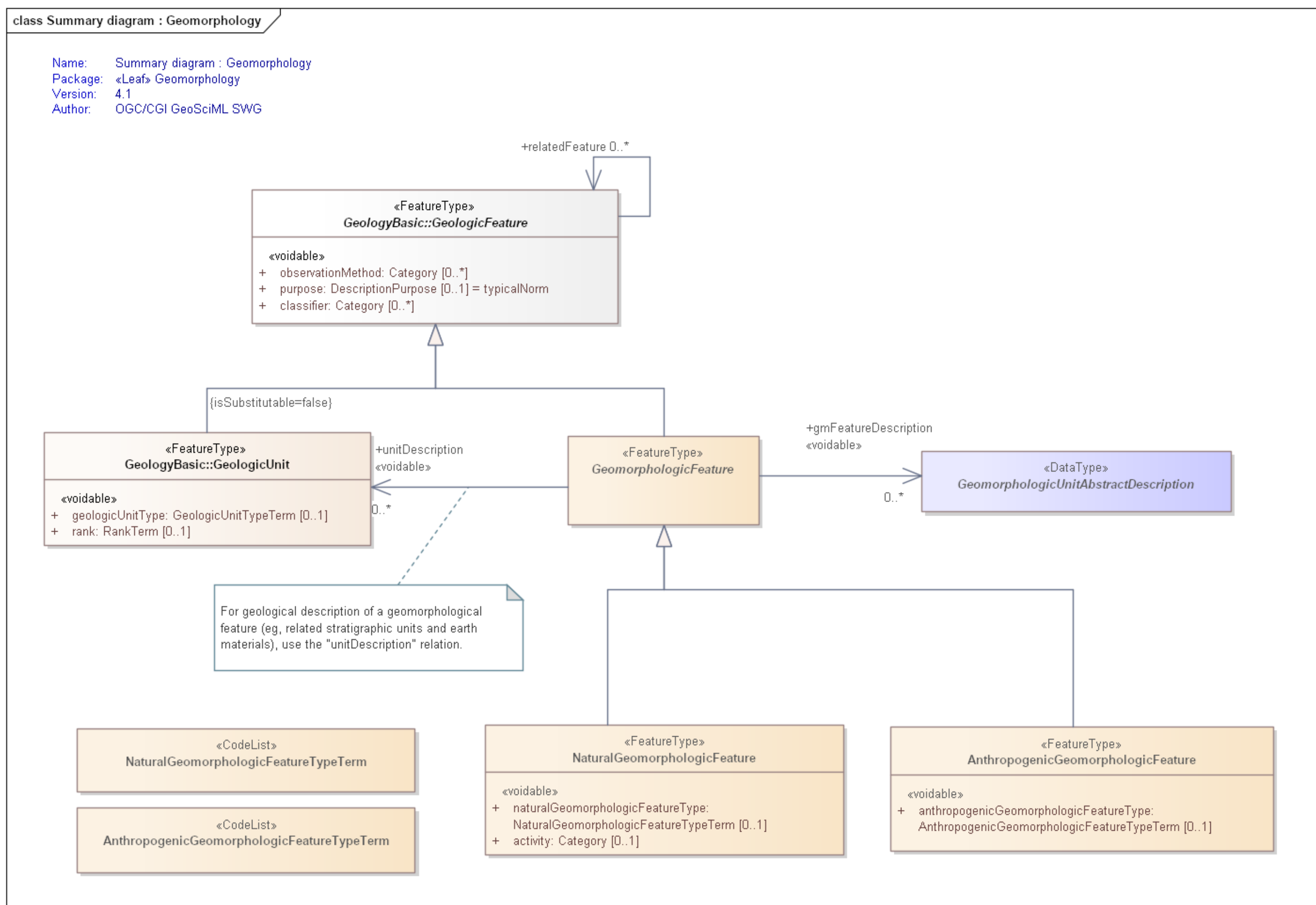


GeologyStructure

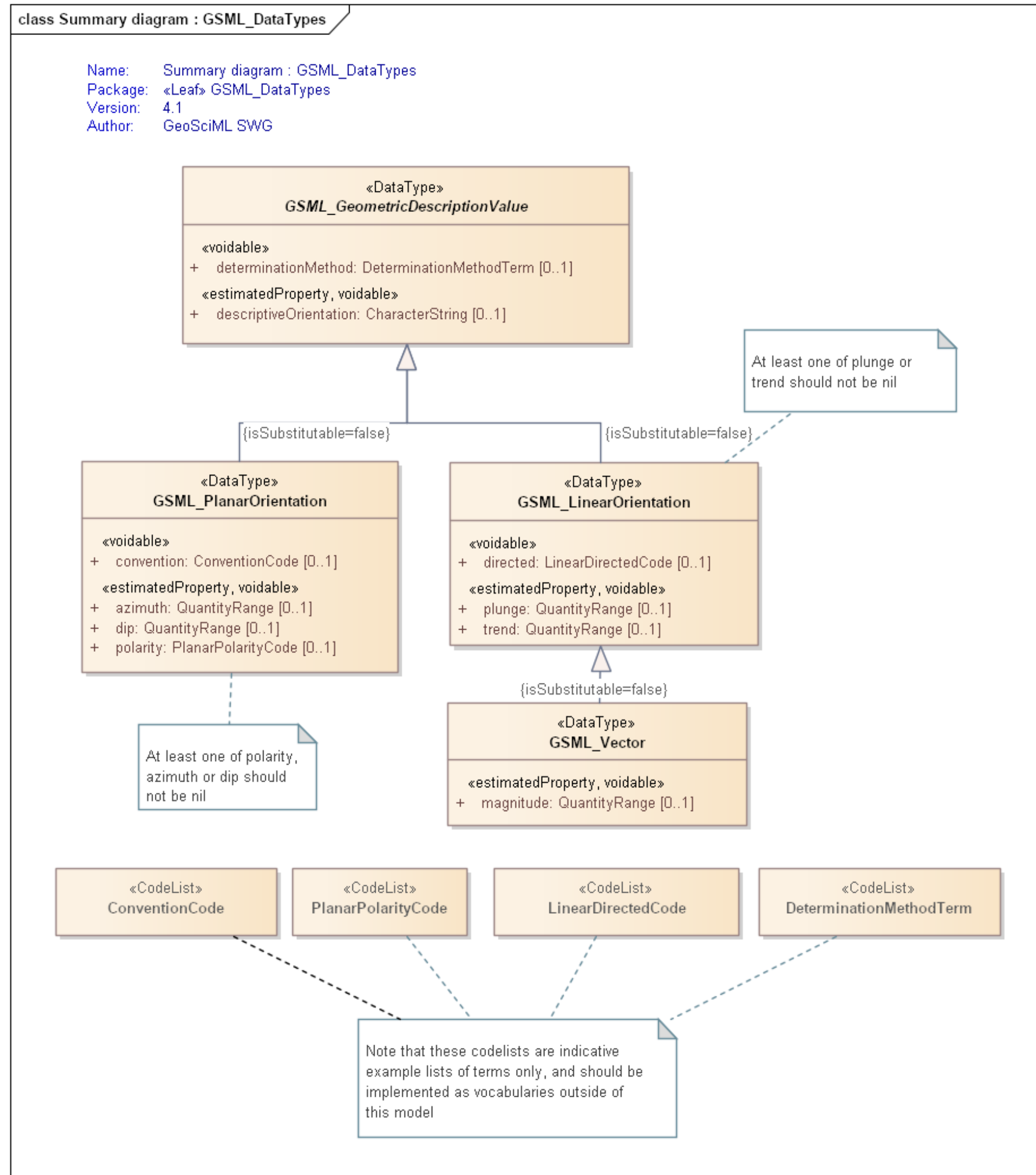




Geomorphology

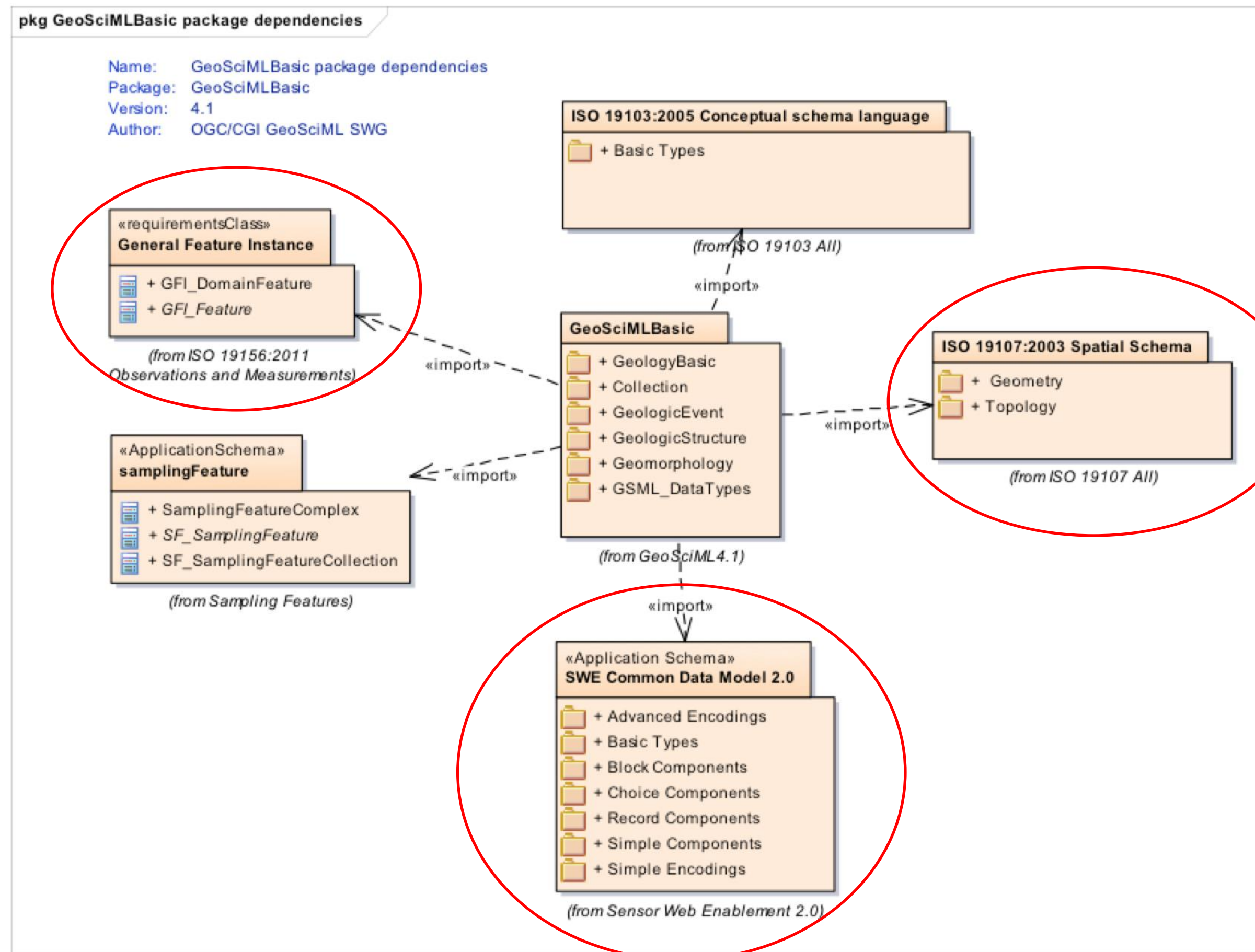


GSML_DataTypes



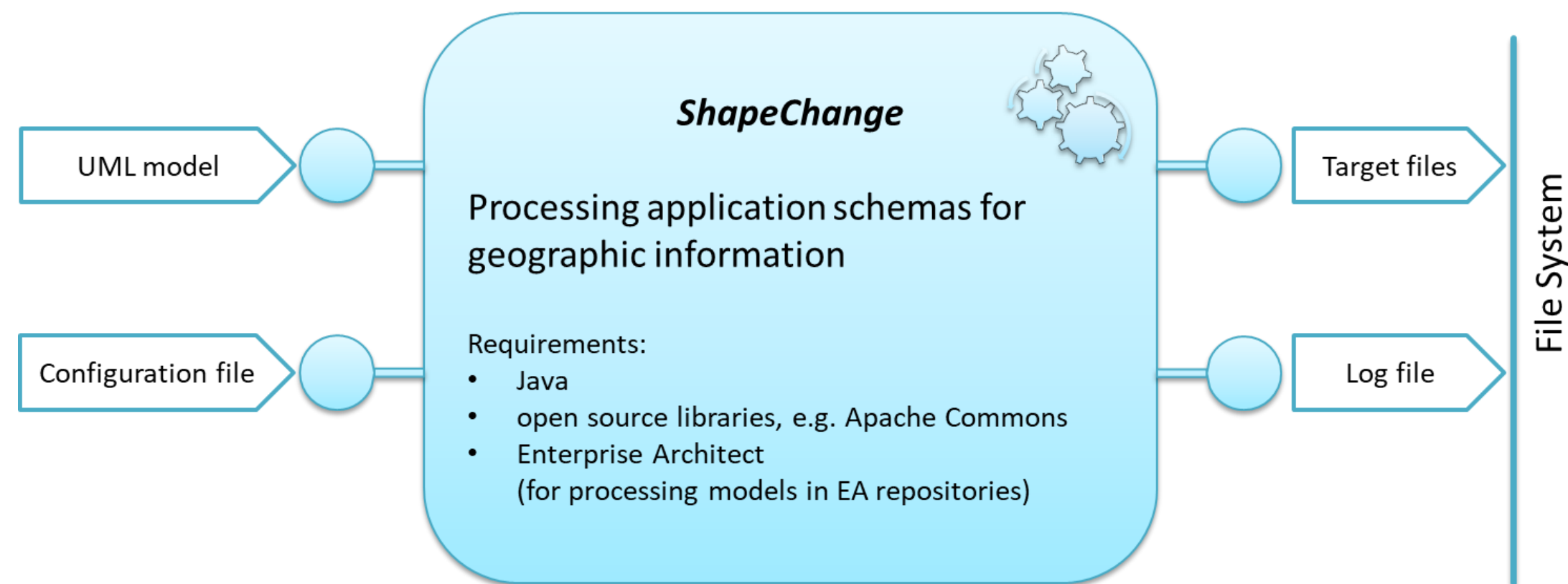
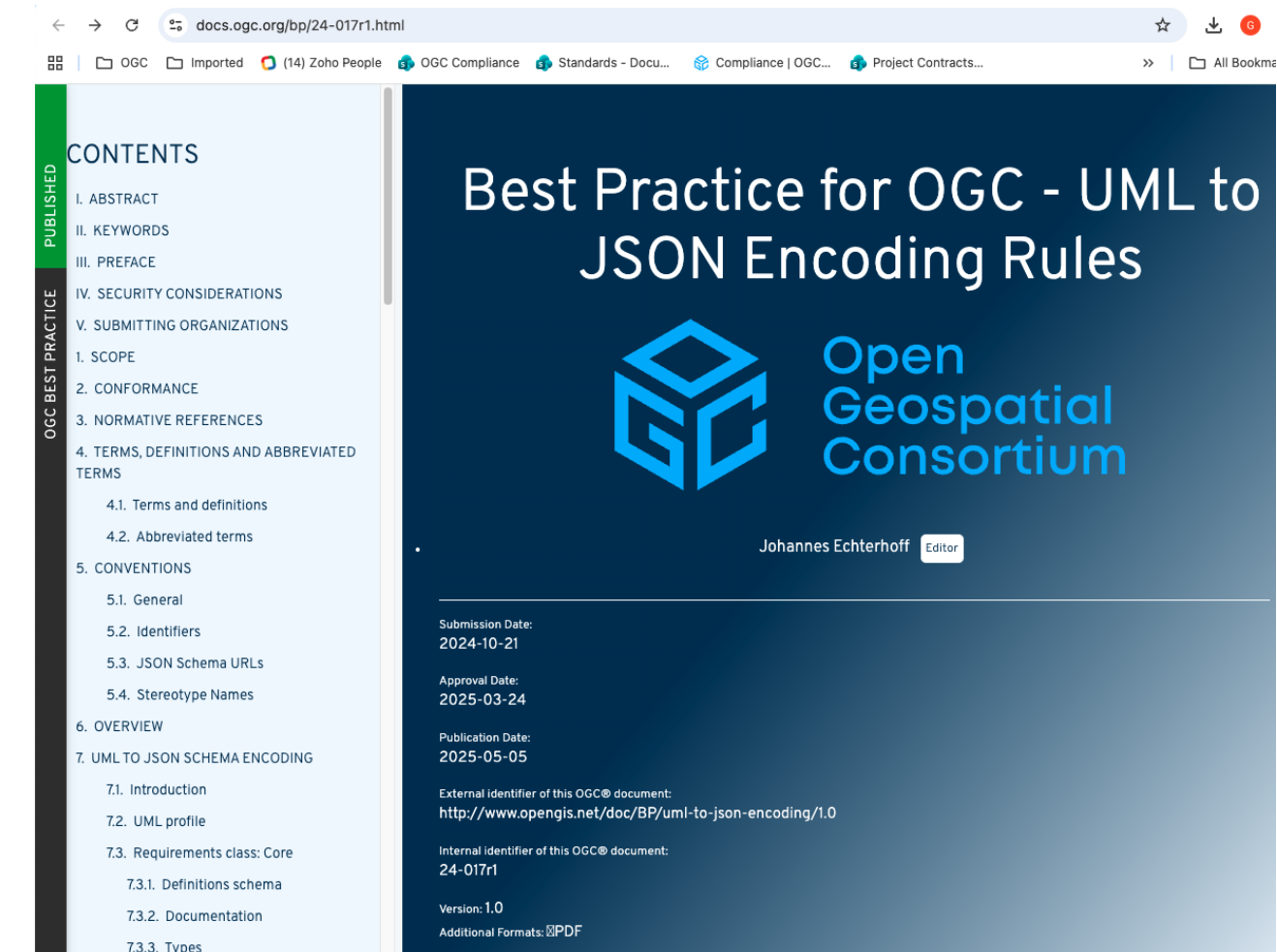
Dependencies and recent developments in the OGC Standards Baseline

- Introduction of JSON-FG creates opportunities for representation of Features and Geometry
- Recent release of SWE Common 3.0 (JSON Schema) creates opportunities



How will the JSON Schema files be developed?

- UML to JSON Encoding Rules
- ShapeChange
 - <https://shapechange.net/>
 - <https://shapechange.github.io/ShapeChange>
- AI (where appropriate or useful)



GeoSciML Vocabularies

- Published by CGI at <https://geosciml.org/resource/>
- Organized into 35 collections of terms
- RDF Turtle files available on [GitHub](#)

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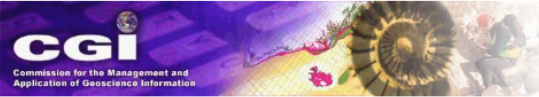
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geosciml.org/resource/

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CGI Home | GeoSciML Home | GeoSciML Wiki | EarthResourceML | Vocabularies

Geoscience Vocabularies for Linked Data

For description of geological features, geological time, mineral occurrences, and mining-related features

Vocabularies for GeoSciML and EarthResourceML

- [Full list of all CGI vocabularies](#)
 - includes links to the CGI vocabulary service, SPARQL endpoints, and RESTful API's for all vocabularies
 - The vocabularies are also published at the [Research Vocabularies Australia Portal](#).
- [Vocabulary files for GeoSciML](#) (TTL)
- [Vocabulary files for the EarthResourceML](#) (TTL)

Vocabularies for Geological Time - the ICS international stratigraphic timescale

- [Geological Timescale service - landing page](#)
- [CGI geological time vocabulary service](#)
- [Vocabulary files for the International Stratigraphic Chart](#) (RDF, TTL, HTML)
- [International Stratigraphic Chart, Interactive version](#)

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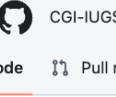
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github.com/CGI-IUGS/cgi-vocabs/tree/master/vocabularies/geosciml

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CGI-IUGS / cgi-vocabs

CodePull requestsActionsProjectsSecurity and qualityInsights

Files

master

Go to file

> .github

> background-onts

> scripts

> vocabularies

> earthresourceml

> geosciml

alterationtype.ttl

boreholedrillingmethod.ttl

compositioncategory.ttl

compoundmaterialconstituen...

consolidationdegree.ttl

contacttype.ttl

conventioncode.ttl

deformationstyle.ttl

descriptionpurpose.ttl

cgi-vocabs / vocabularies / geosciml

Add file

...

vedgell - Move all vocabs back to Vocabs directory for testing action

aef3cbc · 3 years ago

History

This branch is 28 commits behind GeoscienceAustralia/cgi-vocabs:master .

Name	Last commit message	Last commit date
..		
alterationtype.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
boreholedrillingmethod.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
compositioncategory.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
compoundmaterialconstituentpartrole.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
consolidationdegree.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
contacttype.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
conventioncode.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
deformationstyle.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago
descriptionpurpose.ttl	- Move all vocabs back to Vocabs directory for testing action	3 years ago

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cgi.vocabs.ga.gov.au/vocab/

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CGI Vocabularies Register

This is the static register (index) of CGI's vocabularies whose purpose is to present the pe

For machine-readable versions of this vocabulary list, see the [Alternate Profiles](#) page.

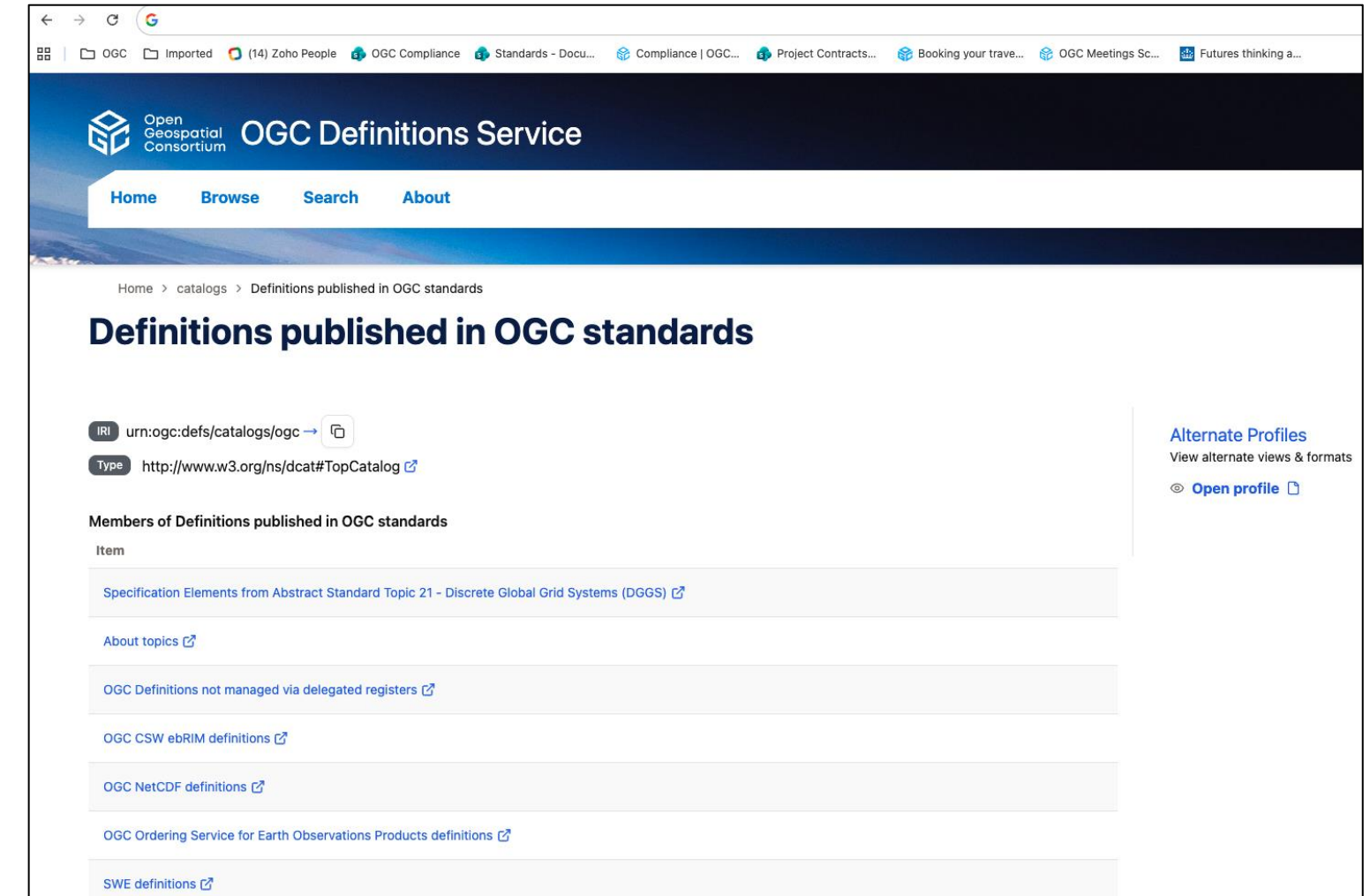
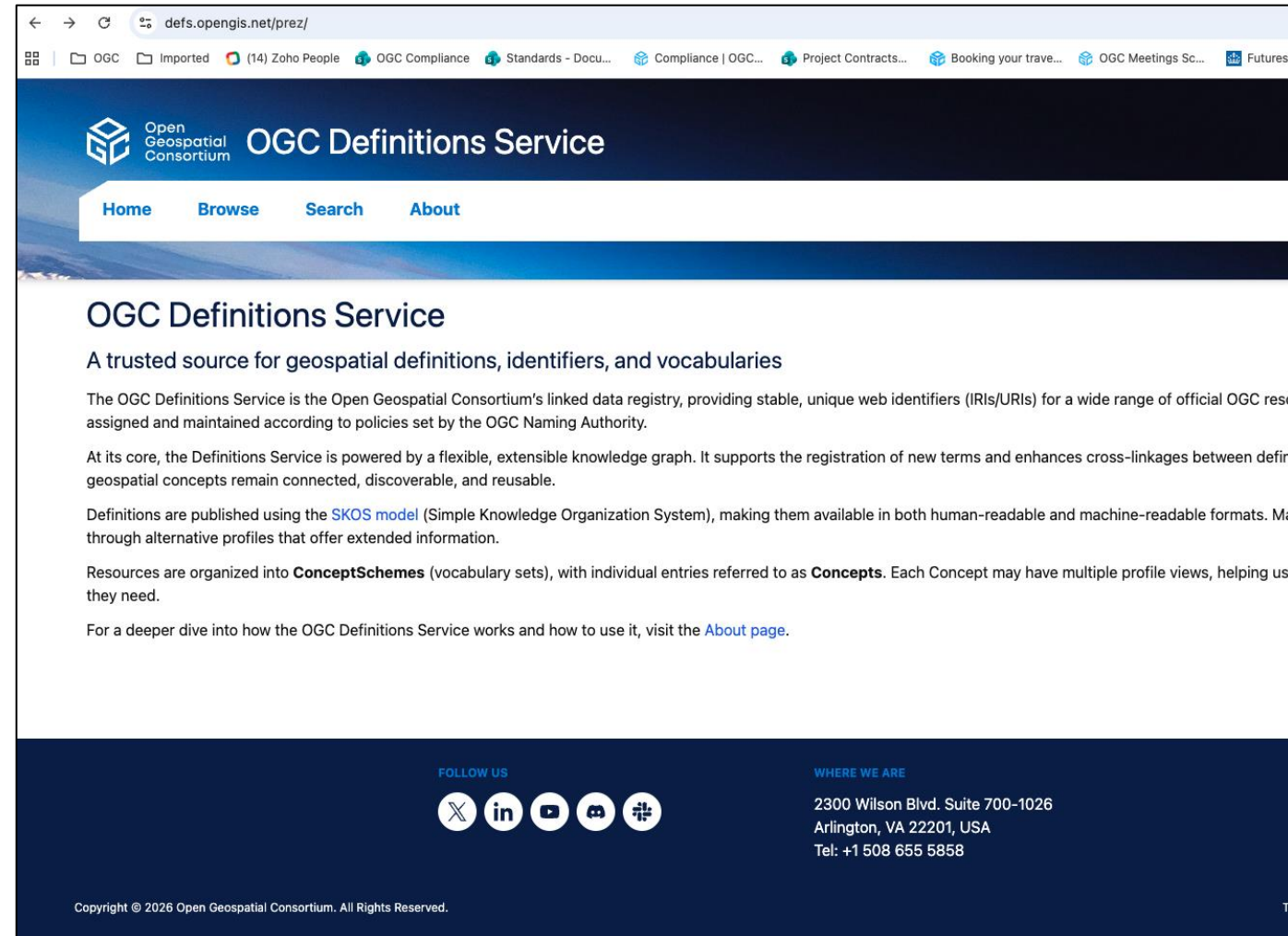
GeoSciML Vocabularies

Collection of terms	Concept Schemes
Alteration Type	v2016.01
Borehole Drilling Method	v2016.01
Composition Category	v2016.01
Compound Material Constituent Part	v2016.01
Consolidation Degree	v2016.01
Contact Type	v2016.01
Convention Code for Strike and Dip Measurements	v2016.01
Deformation Style	v2016.01
Description Purpose	v2016.01
Determination Method Orientation	v2016.01
Event Environment	v2016.01
Event Process	v2016.01
Fault Movement Sense	v2016.01
Fault Movement Type	v2016.01
Fault Type	v2016.01
Feature Observation Method	v2016.01
Foliation Type	v2016.01
Genetic Category	v2016.01
Geologic Unit Morphology	v2016.01
Geologic Unit Part Role	v2016.01
Geologic Unit Type	v2016.01
Lineation Type	v2016.01
Mapped Feature Observation Method	v2016.01
Mapping Frame	v2017.01
Metamorphic Facies	v2016.01
Metamorphic Grade	v2016.01
Particle Aspect Ratio	v2016.01
Particle Shape	v2016.01
Particle Type	v2016.01
Planar Polarity Code	v2017.01
Proportion Term	v2016.01
Simple Lithology	v2016.01
Stratigraphic Rank	v2016.01
Value Qualifier	v2016.01
Vocabulary Relation	v2016.01



Preparation and upload to the OGC Definitions Server (formerly named RAINBOW)

- OGC's linked data registry, providing stable, unique web identifiers (IRIs/URIs) for a wide range of official OGC resources.
- The identifiers are assigned and maintained according to policies set by the OGC Naming Authority (OGC-NA).



How to get involved during the Code Sprint?

- Help write the candidate standard (in metanorma asciidoc)
- Help review the JSON Schema files
 - Compare to UML
 - Create sample JSON documents
 - Validate the sample JSON documents

Contacts during the Code Sprint

- Discord handles
 - Gobe Hobona - @ghobona1
 - Alejandro Villar - @alejandrovillar0321
 - Johannes Echterhoff - @jechterhoff





THANK YOU



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